

Supporting Information for

A Polarizable Cationic Dummy Metal Ion Model

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S.1. Polarizable and non-polarizable Potentials parameter optimizations

S.1.1. Polarizable Force Field Parameter Optimization

Density Functional Theory (DFT) calculations were used to parametrize the potential for the CDA^{pol} model in the presence of water molecules. The potential energy differences of the Al³⁺ ion coordinating with 1-7 water molecules were calculated. Calculations were performed using the Gaussian 16 package and the B3LYP functional combined with the 6-311+g(d,p) basis set for molecules in vacuum. The energy differences were recorded for the metal ion coordinating with different number of water molecules as shown in Figure S.1. This energy profile shows that the minimum energy configuration is when Al³⁺ is coordinating octahedrally with six water molecules. These training data were used to properly tune the CDA parameters to prevent undercoordination and overcoordination geometries and obtain correct coordination numbers during CDA based simulations. In addition to the energy data used for training the force field, charges were assigned to the middle ion and dummy atoms; a partial charge of zero was assigned to the middle ion and a +0.33 charge was assigned to each of the 6 dummy atoms; hence giving a +3 charge for the Al ion. The electrostatic parameters (electronegativity, hardness, and electrostatic shielding) were tuned using the training data. The details of the training data used for tuning the polarizable potential for the octahedral cationic dummy atom coordinating with 1-6 water molecules can be found in section S.2.

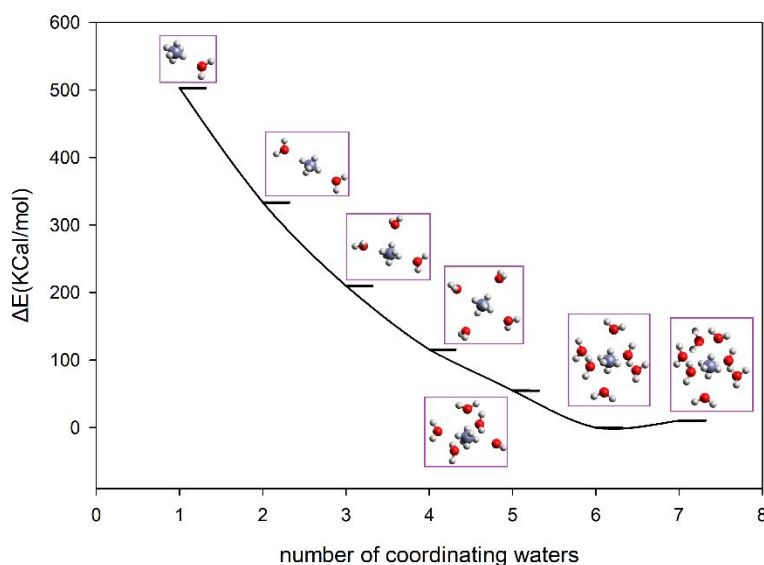


Figure S.1. QM energy profile data used for training the Al³⁺ CDA^{pol} model potential. The data is generated by changing the number of coordinating water molecules from 1 to 7.

The force field is reparametrized to minimize the error function using the QM training data:

$$e_i = \left(\frac{x_{i,QM} - x_{i,R}}{\sigma_i} \right)^2 \quad (\text{S.3})$$

where $x_{i,QM}$ and $x_{i,R}$ are the QM optimized values of the i^{th} entry of the training set, respectively, and σ_i are weight parameters that determine the desired accuracy for individual training data items. After completion of potential optimization, the updated potential was used for thermodynamic integration calculations of an Al^{3+} ion in TIP3P water using the CDA^{pol} .

S.1.2. Lennard-Jones parameter optimization

The 12-6 Lennard-Jones model was parametrized for the Zn^{2+} , Al^{3+} and Zr^{4+} ions within the CDA^{pol} representation. The TIP3P water model was used in all parametrizations and simulations. An octahedral dummy atom model was used for this work which includes one middle ion and six dummy atoms attached to the middle ion (see Figure S.2). As indicated in Table S.1, this octahedral geometry was supported by bond and angle force constants. The interaction between the CDA^{pol} model and the surrounding environment was based on non-bonded interactions (LJ and electrostatic).

Table S.1. Characteristics of the octahedral CDA model where K_b is bond force constant, K_θ is angle force constant, and $U_b = K_b(b - b_0)^2$, $U_\theta = \frac{1}{2}K_\theta(\theta - \theta_0)^2$

Ion	Middle ion charge	Dummy atom charge	$m^M \left(\frac{\text{g}}{\text{mol}} \right)$	$m^D \left(\frac{\text{g}}{\text{mol}} \right)$	$K_b \left(\frac{\text{KCal}}{\text{mol} \cdot \text{\AA}^2} \right)$	$K_\theta \left(\frac{\text{KCal}}{\text{mol} \cdot \text{\AA}^2} \right)$
Zn^{2+}	0	0.33	47.38	3	800	250
Al^{3+}	0	0.5	8.98	3	800	250
Zr^{4+}	0	0.66	73.22	3	800	250

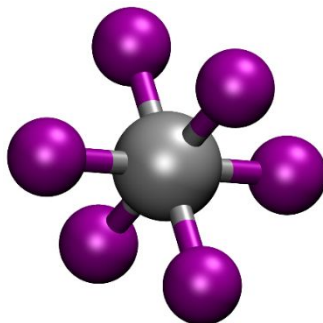


Figure S.2: Octahedral cationic dummy atom. The metal ion in the middle is attached to six dummy atoms in an octahedral geometry.

Two dimensional Lennard-Jones (LJ) parameter scans were performed for the ions. In this protocol, the AMBER LJ parameters $R_{\min}$ and ϵ were changed (see Eq. S.1), while fixed masses were assigned to the middle ion and the dummy atoms.

$$V_{i,j} = \epsilon_{i,j} \left[\left(\frac{R_{\min,i,j}}{r_{i,j}} \right)^{12} - 2 \cdot \left(\frac{R_{\min,i,j}}{r_{i,j}} \right)^6 \right] \quad (\text{S.1})$$

The LJ parameter ϵ was varied from 1 kcal/mol to 3.4 kcal/mol in increments of 0.1, while the $R_{\min}/2$ parameter was scanned from 0.6 Å to 1.0 Å with a 0.1 Å interval. For each LJ parameter pair, the thermodynamic integration (TI) method was utilized to calculate the hydration free energies. The TI method employs a mixed potential of the initial and final states of the system while evolving the systems during the simulation:

$$V(\lambda) = (1 - \lambda)^k \cdot V_0 + (1 - (1 - \lambda)^k) \cdot V_1 \quad (\text{S.2})$$

In the above equation, λ moves between 0 and 1, V_0 is the potential of the initial state and V_1 is the potential of the final state. In this work, the potential was calculated based on linear mixing ($k = 1$).

To start the CDA model simulations, ions were solvated in a TIP3P water box with a length of 20 Å in each dimension. In total, 2736 water molecules were added to the simulation box. Two separate free energy transformations were considered. First, the charges on the CDA ion were removed, and secondly, the LJ interactions between the CDA ion and water molecules were turned off. Our initial investigation showed that different k values and more than three sampling windows did not show significant differences. Therefore, linear mixing was used for all simulations and for each LJ parameter combination, sampling was performed in three windows ($\lambda \in \{0.11270, 0.5, 0.88729\}$). Vacuum calculations were performed in a single window since they converged quite fast. Two separate sets of TI

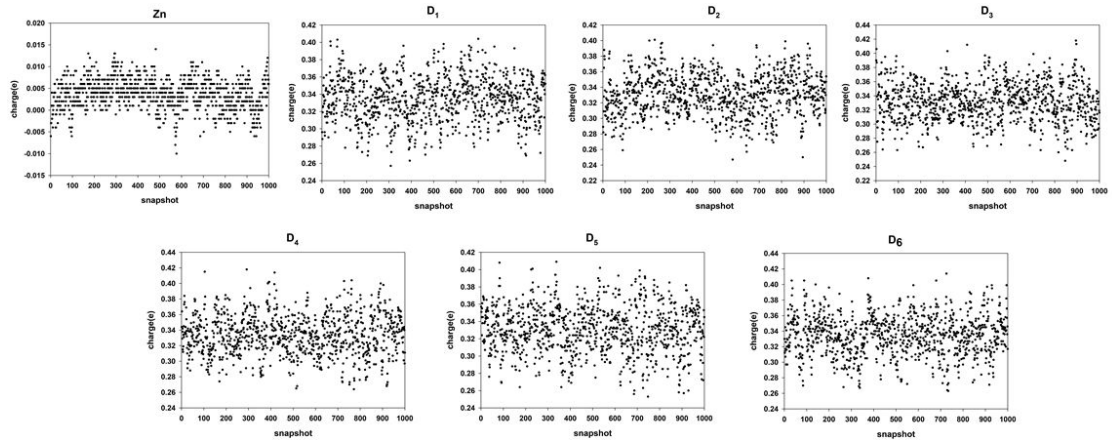
calculations were performed using non-polarizable and polarizable models implemented in the SANDER CPU version of AMBERTools 21. SANDER (Simulated Annealing with NMR-Derived Energy Restraints) is the CPU-based MD engine of Amber. In the first approach, all atoms were modeled with the AMBER force field. In the second approach, the CDA was treated as a polarizable region using the EEM charge equilibration model coupled with TIP3P water molecules. Periodic boundary conditions with the particle mesh Ewald (PME)¹ method was used for all simulations. We chose a timestep of 0.25 fs for this parametrization effort and this was used throughout. Larger timesteps are possible with CDA^{pol} as illustrated by a series of calculations using Zn²⁺ CDA^{pol} model (see Table S.2). The nonbonded interaction cut-off was set to 10 Å for both the AMBER and the polarizable region treated with EEM.

The TI protocol included an initial 25 ps NVT heating of the system from 0 to 300K after geometry optimization, followed by a 50 ps NPT equilibration at 300K. The average values of $dV/d\lambda$ of NPT equilibrations were recorded for HFE calculations. For the NVT heating step, the Langevin algorithm was used to control the temperature with a collision frequency of 2 ps⁻¹. For the NPT ensemble equilibration simulations, Berendsen's barostat with isotropic position scaling was used for pressure control with a pressure relaxation time set to 2 ps.

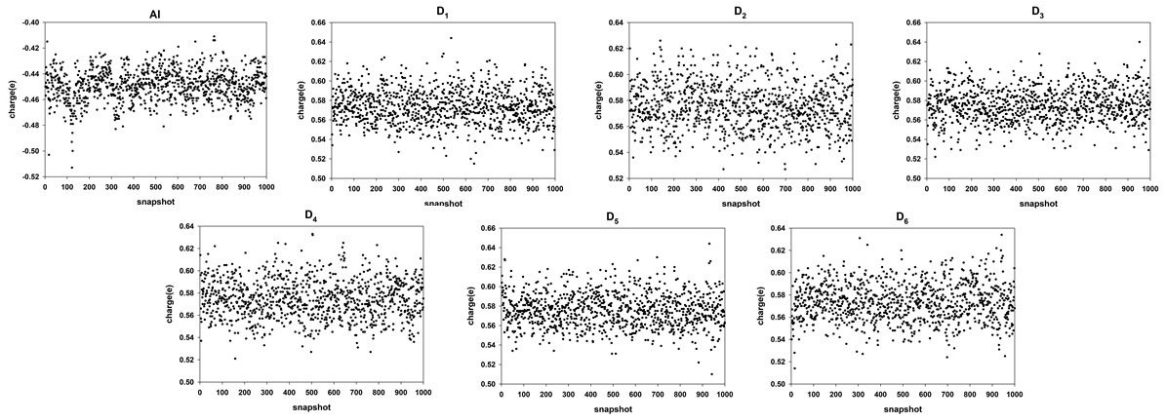
To systematically investigate the effect of the central metal LJ parameter on its HFE and IOD values, a scanning routine was adapted from previous work and applied in this study². The scanning was accomplished by looping over all physical combinations of $R_{\text{min}}/2$ and ϵ values, updating the value in the "frcmod" forcefield modifier file, and then using the tLEaP program to generate a series of initial "prmtop" parameter-topology files³. This step uses the Lorentz–Berthelot combining rules⁴ to mix the modified metal LJ parameters and default water oxygen LJ parameters and finally generate pairwise LJ potentials for the simulations.

S.2. CDA^{pol} charge fluctuations

Zn²⁺ CDA^{pol} charges



Al³⁺ CDA^{pol} charges



Zr⁴⁺ CDA^{pol} charges

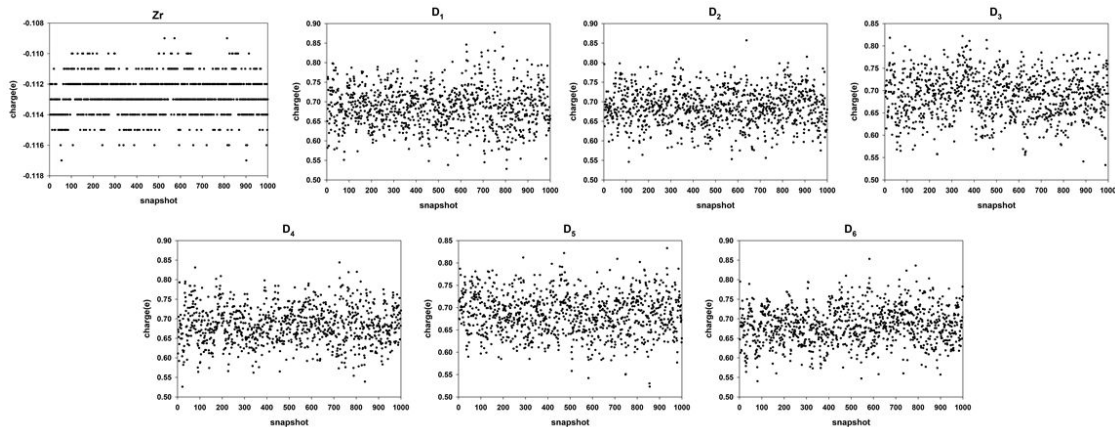


Figure S.3. Charge fluctuations on the middle ion and dummy atoms during equilibration of the solvated CDA^{pol} model

S.3. Dipole moment calculation

The dipole moment of the CDA^{pol} model was calculated with respect to the molecule's center of mass as follows:

$$x_c = \left(\frac{1}{M}\right) \cdot \sum_{i=1}^n (m_i \times x_i) ; y_c = \left(\frac{1}{M}\right) \cdot \sum_{i=1}^n (m_i \times y_i) ; z_c = \left(\frac{1}{M}\right) \cdot \sum_{i=1}^n (m_i \times z_i)$$

$$P_x = \sum_{i=1}^n [q_i \times (x_i - x_c)] ; P_y = \sum_{i=1}^n [q_i \times (y_i - y_c)]$$

$$P_z = \sum_{i=1}^n [q_i \times (z_i - z_c)]$$

$$P = \sqrt{P_x^2 + P_y^2 + P_z^2}$$

where x_i , y_i , and z_i are atom coordinates, m_i are the atomic masses, M is the total mass of the molecule, (x_c, y_c, z_c) are the center of mass coordinates, q_i are atom charges, and P is the dipole moment.

Table S.2. HFE, IOD, and CN for hydration of Zn²⁺ CDA^{pol} LJ parameters at time steps bigger than 0.25fs

Time step	IOD (Å)	CN	HFE (Kcal/mol)
0.5 fs	2.04	6.3	-464.8
1 fs	2.04	6.3	-465.6
1.5 fs	2.12	6.3	-465.5
2 fs	2.05	6.1	-465.8

S.4. Electronegativity equalization method

The EEM is a fluctuating charge model which relies on the principle that charges should be distributed on atoms to satisfy constraints for both the net system charge and the equalized atom electronegativities. Let atomic charges be $\mathbf{q} = (q_1, q_2, \dots, q_n)$ and the positions be $\mathbf{r} = (r_1, r_2, \dots, r_n)$ where $q_i \in \mathbf{R}$ and $r_i \in \mathbf{R}^3$. Based on Sanderson's Electronegativity Equalization Principle, the electronegativity of all atoms needs to be equalized.

$$\epsilon_1 = \epsilon_2 = \dots = \epsilon_i = \underline{\epsilon}$$

where ϵ_i is the electronegativity of atom i and $\underline{\epsilon}$ is the average molecular electronegativity. The other constraint forces the sum of the atomic charges to be equal to the given net system charge:

$$\sum_{i=1}^n q_i = q_{net}$$

The constraints and the parametrized interatomic interactions can be merged into the following linear equation where the charges $\mathbf{q}$ are the solution of:

$$\begin{bmatrix} \mathbf{H} & \mathbf{1}_n \\ \mathbf{1}_n^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \underline{\epsilon} \end{bmatrix} = \begin{bmatrix} -\boldsymbol{\chi} \\ q_{net} \end{bmatrix}$$

Where, $\boldsymbol{\chi}$ is an n by 1 vector of atomic electronegativities and the H_{jj} values, i.e., individual elements of the n by n $\mathbf{H}$ matrix, are defined as $\delta_{ij}\eta_i + (1 - \delta_{ij}) \cdot F_{ij}$, where δ_{ij} is the Kronecker delta operator, η_i is the idempotential, and $F_{i,j}$ is defined as

$$F_{i,j} = \begin{cases} \frac{1}{\sqrt[3]{r_{ij}^3 + \gamma_{ij}^3}}, & r_{ij} \leq r_{\text{nonb}} \\ 0, & \text{otherwise} \end{cases}$$

In the equations above, $r_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$ is the distance between atoms i and j , $\gamma_{ij} = \sqrt{\gamma_i \cdot \gamma_j}$ is a pairwise *shielding term* tuned for element types of atoms i and j to avoid unbounded electrostatic energy at short distances, and $\underline{\epsilon}$ is the dielectric constant of the medium.

References

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